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IOS+ Engineering Body — Document 4

Gate 530 Compliance Dimension Matrix: UCO-Native Architecture

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Engineering Body Cross-Reference Index

EB Doc 2:	Cryptographic Audit Framework — Ed25519/JCS Signing, Key Lifecycle Management
EB Doc 3:	COS+ Schema Reference (including Amendment v1.1) — UCO Node Tables, Tenant Profiles, Agency Registry
EB Doc 4:	THIS DOCUMENT — Gate 530 Compliance Dimension Matrix: UCO-Native Architecture
EB Doc 5:	RAG Vault Architecture — Sector Partitions, UCO Node Vector Embeddings
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1 — Executive Summary

Gate 530 is the compliance enforcement layer within the IOS+ Yellow Brick Road (YBR) architecture, operating at Layer 4 (L4). Prior conceptual documentation described Gate 530 as evaluating against an abstract "127-dimension, 9-group" compliance matrix. **This document supersedes that conceptual framing and defines the production architecture.**

► Production Architecture — Definitive Statement

Gate 530 evaluates against the **Universal Compliance Ontology (UCO)** — a live knowledge graph of **350 compliance obligation nodes** derived from the SMEPro Universal Compliance Decoding Matrix. These 350 UCO nodes *are* the compliance dimensions. The abstract dimension concept is retired in all production contexts.

Each UCO node encodes a specific regulatory obligation comprising:

- A governing agency and its jurisdictional scope
- A named regulation with its CFR/USC citation

- A penalty structure (monetary amounts, criminal exposure, license consequences)
- A risk weight (integer 1–10; floor is 5 in production dataset)
- A deterministic policy action: **BLOCK**, **APPROVE**, or **ESCALATE**
- An enforcement type classification (Criminal, Civil Monetary Penalty, Administrative, License/Certificate, Injunctive, Warning/Notice)

Gate 530's five-function operational mandate is:

13. Resolve the tenant's industry profile via their registered NAICS codes
14. Identify the applicable UCO node subset for that tenant's profile
15. Evaluate each applicable node against the inbound query context
16. Aggregate individual node evaluations to a single gate decision (PASS / BLOCK / QUARANTINE)
17. Emit a cryptographically signed Gate Decision Record referencing all evaluated UCO_NODE_IDs

⚠ Critical — Fail-Closed Architecture

Gate 530 operates **fail-closed**. Any degraded state (timeout, database unavailability, UCO graph error) results in a default response of **BLOCK**. No query passes compliance evaluation in a degraded operational state. This is non-negotiable and may not be overridden by configuration.

2 — UCO Architecture Overview

2.1 From Abstract Dimensions to Live Compliance Nodes

The Universal Compliance Ontology replaces the abstract dimension concept in all production contexts. Each UCO node is a **first-class data object** persisted in COS+ (table: `uco_nodes`; schema defined in EB Doc 3 including Amendment v1.1). The 350-node graph is organized into two structural layers:

- **19 industry sector layers** — prefixed UCO-ENR-1xxx through UCO-MGT-19xx — covering all primary NAICS-aligned industry groupings
- **1 universal cross-cutting layer** — prefixed UCO-XSC-5xxx — comprising 19 nodes that apply to *all* tenants regardless of sector classification

The sector prefix architecture provides deterministic routing: any UCO_NODE_ID can be immediately mapped to its sector, risk profile, and governing agency without a database lookup, enabling pre-fetch optimization at the L3 ontological mapping layer.

2.2 UCO Node Anatomy

Each UCO node carries **30 attributes** organized across two functional groups:

Group A — Regulatory Identity (Columns 1-20)

Col	Attribute Name	Description / Example Values
1	Broad Industry	Top-level industry grouping (e.g., "Energy", "Healthcare")
2	Industry Subtype	Narrowed classification within broad industry
3	Specific Activity / Process	The exact regulated activity or business process
4	Jurisdiction Level	Federal, State, Local, or Multi-jurisdictional
5	Governing Agency	Regulatory body with enforcement authority

Col	Attribute Name	Description / Example Values
		(e.g., BSEE/BOEM, CMS, OCC)
6	Regulation Name	Full formal regulation name
7	CFR / USC Citation	Code of Federal Regulations or US Code citation (e.g., 30 CFR 250)
8	Report / Form Name	Associated regulatory filing form name
9	Form Number / Code	Official form identifier
10	Filing Frequency	Annual, Quarterly, Monthly, On-Event, Continuous
11	Key Due Dates / Deadlines	Statutory or regulatory filing deadlines
12	Business Segment	Functional business area subject to regulation
13	Penalties / Consequences	Monetary penalties, criminal exposure, license actions (e.g., "\$40K/day")
14	CIP Code	Classification of Instructional Programs code (education crosswalk)
15	SIC Code	Standard Industrial Classification code (legacy system crosswalk)
16	NAICS Code	North American Industry Classification System code (primary routing key)
17	SOC Code	Standard Occupational Classification code
18	ISIC Code	International Standard Industrial Classification code

Col	Attribute Name	Description / Example Values
19	HS / HTS Code	Harmonized System / Harmonized Tariff Schedule code (trade crosswalk)
20	Notes	Supplementary regulatory context, amendment notes, state-specific variations

Group B — COS+ Engine Metadata (Columns 21-30)

Col	Attribute Name	Description / Example Values
21	UCO_NODE_ID	Primary key — unique node identifier (e.g., UCO-ENR-1001, UCO-XSC-5002)
22	ONTOLOGY_LEVEL	All production nodes: "L2: Regulations and Rules"
23	COMPLIANCE_CHAIN_REF	Full crosswalk chain string (e.g., "CIP:15.0903 → NAICS:211130 → Federal → BSEE/BOEM → Outer Continental Shelf Lands Act")
24	OPERATING_SEGMENT	Business segment label for query routing context matching
25	RESPONSIBLE_ROLE	C-suite role accountable for compliance (e.g., CCO/General Counsel, CFO/Controller)
26	ENFORCEMENT_TYPE	Criminal Civil Monetary Penalty Administrative License/Certificate Injunctive Warning/Notice
27	RISK_WEIGHT	Integer 5-10; 10 = highest risk; floor is 5 in production dataset

Col	Attribute Name	Description / Example Values
28	YBR_GATE	All production nodes: "Gate 530: Compliance Check"
29	POLICY_ACTION	BLOCK APPROVE ESCALATE — deterministic gate action signal
30	LAST_UPDATED	Timestamp of most recent regulatory content update to this node

2.3 UCO Node Distribution by Sector

The following table defines the complete 350-node distribution across all 20 sector classifications (19 industry sectors + 1 universal cross-cutting layer). Node counts are production-validated against the SMEPro Universal Compliance Decoding Matrix.

Sector #	Sector Name	UCO Node Prefix	Node Count	Risk Tier	Primary Policy Action
01	Energy	UCO-ENR-1xxx	54	CRITICAL	BLOCK
02	Healthcare	UCO-HCR-2xxx	36	CRITICAL	BLOCK
03	Finance & Banking	UCO-FIN-3xxx	30	CRITICAL	BLOCK
04	Food, Drug & Agriculture	UCO-FDA-4xxx	16	HIGH	BLOCK
05	Manufacturing & Transport	UCO-MFG-5xxx	27	HIGH	BLOCK
06	Telecom, Environment & Defense	UCO-TEL-6xxx	20	HIGH	BLOCK
07	Insurance	UCO-INS-7xxx	35	CRITICAL	BLOCK

Sector #	Sector Name	UCO Node Prefix	Node Count	Risk Tier	Primary Policy Action
08	Real Estate & Construction	UCO-REC-8xxx	10	HIGH	BLOCK
09	Agriculture	UCO-AGR-9xxx	8	MEDIUM	APPROVE
10	Mining	UCO-MIN-10xx	5	HIGH	BLOCK
11	Wholesale & Retail	UCO-WRT-11xx	15	HIGH	BLOCK
12	Professional Services	UCO-PRO-12xx	13	HIGH	APPROVE
13	Education	UCO-EDU-13xx	10	CRITICAL	APPROVE
14	Arts & Entertainment	UCO-ART-14xx	9	HIGH	BLOCK
15	Accommodation & Food Services	UCO-ACF-15xx	10	MEDIUM	ESCALATE
16	Administrative & Waste Services	UCO-ADM-16xx	9	HIGH	APPROVE
17	Other Services	UCO-OTH-17xx	9	HIGH	BLOCK
18	Public Administration	UCO-PUB-18xx	9	CRITICAL	APPROVE
19	Management of Companies	UCO-MGT-19xx	6	HIGH	APPROVE
XSC	Cross-Cutting (Universal)	UCO-XSC-5xxx	19	CRITICAL	BLOCK
ALL	TOTAL — Production UCO	20 Sector Layers	350	—	—

Sector #	Sector Name	UCO Node Prefix	Node Count	Risk Tier	Primary Policy Action
	Graph				

► **Note — Sector 01 (Energy) Dominance**

Energy carries 54 nodes (15.4% of total), reflecting the extraordinary density of federal and state regulatory regimes governing oil & gas, nuclear, pipeline, offshore drilling, and environmental compliance. Healthcare (36) and Insurance (35) follow as the next most regulated sectors.

3 — Risk Weight and Threshold Architecture

3.1 Risk Weight Distribution

Risk weights in the production 350-node dataset are distributed as follows. Risk weight values are integers in the range 5–10; no production UCO node carries a risk weight below 5 (the matrix floor). This distribution reflects the deliberate exclusion of trivially low-risk regulatory obligations from the production compliance graph.

Risk Weight	Node Count	% of Total	Gate Decision Implication
10 — Maximum	90	25.7%	Automatic BLOCK — no override permitted; criminally enforced or catastrophically

Risk Weight	Node Count	% of Total	Gate Decision Implication
			penalized regulations
9 — Critical	38	10.9%	BLOCK unless explicit verified exemption is recorded in tenant compliance attestation
8 — High	64	18.3%	BLOCK or QUARANTINE per policy_action field value
7 — Elevated	33	9.4%	ESCALATE for compliance officer review
6 — Moderate-High	17	4.9%	ESCALATE with enhanced monitoring and audit logging
5 — Baseline (Floor)	108	30.9%	APPROVE with full audit record — lowest actionable risk tier
ALL	350	100.0%	Floor = 5 Ceiling = 10 No nodes below risk weight 5

⚠ Architectural Note — Risk Weight 5 Node Count

The 108 risk-weight-5 nodes (30.9%) represent the largest single cohort. These nodes correspond primarily to administrative reporting obligations with APPROVE policy actions. Despite their APPROVE outcome, they generate full audit records and contribute to the tenant's compliance posture score. They cannot be skipped during evaluation.

3.2 Risk Weight to Gate Decision Mapping

Gate 530 applies a deterministic, ordered decision algorithm. The mapping is evaluated in priority order; the first matching condition determines the gate decision. **The algorithm is not configurable by tenant or administrator.**

Condition	Risk Weight	Policy Action	Gate Decision	Override Permitted
Rule 1 — Absolute Block	10	Any	BLOCK	None
Rule 2 — High-Risk Block	8-9	BLOCK	BLOCK	None
Rule 3 — High-Risk Quarantine	8-9	ESCALATE	QUARANTINE	Human review required within 4-hour SLA
Rule 4 — Elevated Quarantine	5-7	ESCALATE	QUARANTINE	Human review required within 24-hour SLA
Rule 5 — Approved with Audit	5-7	APPROVE	PASS	N/A — full audit record generated
Rule 6 — Fail-Closed Override	Any	Any	BLOCK	Applied if ANY single node triggers BLOCK (fail-closed aggregation)

Example regulatory anchors for Rule 1 (Risk Weight 10 — Absolute Block):

- **UC0-MIN-10xx** — Coal Mining Safety (MSHA): \$214,022/day civil penalties + criminal prosecution
- **UC0-TEL-1164** — Wireless Spectrum Licensing (FCC): \$10M+ administrative penalties
- **UC0-ART-14xx** — Casino Gaming Licensing (State Gaming Boards): Criminal enforcement, permanent license revocation

- [UC0-XSC-5002](#) — FLSA Wage & Hour Compliance (DOL-WHD): Criminal penalties + back pay liability
- [UC0-XSC-5006](#) — IRS Trust Fund Recovery Penalty: 100% penalty of unpaid tax + criminal prosecution

3.3 Fail-Closed Semantics

Gate 530 operates with strict fail-closed semantics throughout its evaluation lifecycle. The fail-closed guarantee extends beyond the risk weight decision algorithm:

- **Database unavailability:** If [uco_nodes](#) cannot be queried (connection pool exhaustion, failover in progress), Gate 530 returns BLOCK immediately without waiting for timeout
- **Tenant profile resolution failure:** If [tenant_naics_profiles](#) cannot be read, the tenant is treated as activating ALL 350 nodes, and the decision is BLOCK
- **Signing service unavailability:** If the Ed25519 signing service (EB Doc 2) cannot be reached, the Gate Decision Record cannot be completed, and the query is BLOCKED pending a signed record
- **Evaluation timeout:** If the total evaluation exceeds the 50ms L4 budget, any pending node evaluations are resolved as BLOCK by default before aggregation
- **Partial UCO graph:** If fewer UCO nodes are loaded than the tenant profile expects, the missing nodes are treated as BLOCK-triggering pending graph recovery

⚠ Architectural Invariant

The fail-closed guarantee is an architectural invariant, not a configurable policy. It cannot be modified by tenant configuration, administrator override, or runtime flags. Any pull request modifying fail-closed behavior requires Engineering Body CCO sign-off and full security review.

4 — Policy Action State Machine

4.1 Policy Action Distribution

The three policy action values are distributed across the 350-node production dataset as follows:

Policy Action	Node Count	% of Total	Aggregate Gate Effect
BLOCK	192	54.9%	Majority of UCO nodes enforce BLOCK — any single trigger produces BLOCK for the entire query
APPROVE	108	30.9%	Access permitted with full audit logging; does not override a concurrent BLOCK
ESCALATE	50	14.3%	Query enters QUARANTINE state; compliance officer review required per tier SLA
TOTAL	350	100.0%	—

4.2 Three-State Evaluation — Detailed Semantics

BLOCK State

Content contains data subject to a regulation where access is prohibited without verified compliance attestation on record in the tenant's compliance profile. The

BLOCK state is non-negotiable at the Gate 530 layer. Compliance attestation (if applicable) must be pre-recorded in `tenant_compliance_attestations` before any re-evaluation is triggered. BLOCK records are appended to the immutable audit log with full `UCO_NODE_ID` attribution.

APPROVE State

The regulation applies to the tenant's context, but access is permitted with audit logging. APPROVE does not mean the regulation is inapplicable — it means the regulation's compliance posture is satisfactory for read access in the current context. Full audit records are generated identifying all APPROVE-triggered nodes, enabling post-hoc compliance reporting and regulatory examination response.

ESCALATE State (→ QUARANTINE)

The regulation applies and the query context raises sufficient risk that a human compliance officer review is required before access may be permitted. The query is placed in QUARANTINE state. The tenant receives a QUARANTINE response code (not an error). A compliance workflow event is generated and routed to the responsible role(s) identified in the triggering node's `RESPONSIBLE_ROLE` attribute. Access is not permitted until the compliance officer records a disposition in the compliance review queue.

4.3 Escalation Tiers

ESCALATE-policy nodes are further classified into escalation tiers based on their `ENFORCEMENT_TYPE`, which determines the urgency, routing, and SLA of the human review requirement:

Enforcement Type	Tier	SLA	Routing	Trigger Example
Criminal	Tier 1	1 hour	Legal Counsel + CCO/General Counsel notification required	UCO-XSC-5004 (OSHA General Duty Clause)
Injunctive	Tier 1	1 hour	Legal Counsel notification + CRO/Risk	UCO-XSC-5001 (Title VII / ADA)

Enforcement Type	Tier	SLA	Routing	Trigger Example
			Manager	Injunctive Relief)
Civil Monetary Penalty	Tier 2	4 hours	Compliance Officer review — CFO/Controller + CCO/General Counsel	UCO-XSC-5008 (State Data Breach Notification)
License / Certificate	Tier 2	4 hours	Compliance Officer review — COO/VP Ops + CRO/Risk Manager	UCO-ENR-1039 (NRC Nuclear Operating License)
Administrative	Tier 3	24 hours	Standard compliance review — COO/VP Ops + CCO/General Counsel	UCO-XSC-5003 (FMLA Leave Administration)

► SLA Enforcement Note

Escalation SLAs are enforced at the application layer via scheduled compliance queue monitoring. If an escalation item remains unresolved past its SLA, an automated escalation notice is generated to the next tier's responsible role. SLA breach itself does not automatically convert QUARANTINE to BLOCK, but it triggers an executive notification and is recorded as a compliance posture event.

5 — Enforcement Type Classification

5.1 The Six Enforcement Types

Gate 530 recognizes six enforcement type classifications, derived from the actual regulatory structure of the 350-node dataset. Enforcement type is a first-class attribute in the UCO node schema (Group B, Column 26) and directly drives escalation tier routing.

Enforcement Type	Node Count	% of Total	Severity	Gate Impact
Civil Monetary Penalty	125	35.7%	High	Risk-weight-dependent BLOCK or APPROVE; CMP amounts range from hundreds to millions per day
Criminal	79	22.6%	Maximum	Always Tier 1 escalated when ESCALATE; risk weight 10 = automatic BLOCK; personal liability exposure for responsible officers
Administrative	70	20.0%	Moderate	Primarily APPROVE with audit logging; lowest urgency in escalation tier hierarchy (Tier 3)
License / Certificate	57	16.3%	High	BLOCK for missing or expired license situations; Tier 2 escalation when

Enforcement Type	Node Count	% of Total	Severity	Gate Impact
				ESCALATE policy
Injunctive	14	4.0%	High	ESCALATE with Tier 1 legal review; court-ordered compliance exposure; includes consent decree scenarios
Warning / Notice	5	1.4%	Low	APPROVE with enhanced monitoring; lowest gate impact; regulatory notice regimes with no immediate penalty
TOTAL	350	100.0%	Civil Monetary Penalty dominates (35.7%); Criminal exposure present in 22.6% of nodes	

5.2 Enforcement Type to Responsible Role Mapping

Each UCO node's RESPONSIBLE_ROLE attribute (Group B, Column 25) encodes the C-suite accountable roles. The mapping from enforcement type to responsible role is derived from the actual dataset and reflects standard corporate governance accountability structures:

Enforcement Type	Primary Responsible Role	Secondary Responsible Role	Rationale
Criminal	CCO / General Counsel	CISO / CTO	Criminal exposure requires legal counsel as primary;

Enforcement Type	Primary Responsible Role	Secondary Responsible Role	Rationale
			CISO/CTO for data-related criminal matters
Civil Monetary Penalty	CFO / Controller	CCO / General Counsel	Financial exposure is CFO-owned; legal counsel required for penalty negotiations
Administrative	COO / VP Operations	CCO / General Counsel	Operational compliance is COO-owned; counsel for regulatory interface
License / Certificate	COO / VP Operations	CRO / Risk Manager	License maintenance is operations-owned; risk manager for license risk exposure assessment
Injunctive	CCO / General Counsel	CRO / Risk Manager	Court-ordered compliance is counsel-primary; risk manager for consent decree monitoring
Warning / Notice	COO / VP Operations	—	Warning regimes are operational matters; no secondary escalation required

6 — Compliance Chain Routing Architecture

6.1 The Compliance Chain

Each UCO node carries a **COMPLIANCE_CHAIN_REF** string (Group B, Column 23) that encodes the complete regulatory lineage from classification code through jurisdiction to agency and regulation. This chain is the canonical audit trace of how a given regulatory obligation was identified as applicable to a tenant.

Chain Format:

```
CIP:{code} → NAICS:{code} → {JurisdictionLevel} → {Agency} →  
{RegulationName}
```

Production Examples:

```
CIP:15.0903 → NAICS:211130 → Federal → BSEE/BOEM → Outer Continental  
Shelf Lands Act CIP:51.3801 → NAICS:622110 → Federal → CMS → Medicare  
Conditions of Participation (42 CFR 482) CIP:52.0201 → NAICS:522110 → Federal  
→ OCC → Community Reinvestment Act (12 CFR 25) CIP:13.1201 →  
NAICS:611310 → Federal → DOE → FERPA (34 CFR Part 99)  
NAICS:UNIVERSAL → ALL → Federal → DOL-WHD → FLSA (29 USC §201)
```

6.2 Chain Resolution at L3 (Ontological Mapping)

At YBR Layer 3 (Ontological Mapping), the query context is resolved to a compliance chain before the evaluation request reaches Gate 530 at L4. The L3 resolver executes the following sequence:

18. Extract entity types from the query's ontological graph (product of L2 processing)
19. Look up the tenant's NAICS codes in `tenant_naics_profiles` using `tenant_id` as the primary key (P99 < 2ms)
20. Resolve applicable `naics_decoder` entries → identify candidate industry sectors
21. Fetch `compliance_chains` rows WHERE `naics_code` IN (tenant's registered NAICS codes) OR `is_cross_cutting` = TRUE
22. Return the candidate UCO node list to Gate 530 via the L4 IPC channel (Unix domain socket, defined in EB Doc 1)

The L3 pre-resolution step ensures Gate 530 receives a pre-filtered candidate set rather than evaluating all 350 nodes against every query. However, Gate 530 independently validates the candidate set against the live `uco_nodes` table — L3 pre-filtering is a performance optimization, not a trust boundary.

6.3 Six-System Code Crosswalk

The `code_crosswalk` table (defined in EB Doc 3 §6.5) maintains bidirectional mappings between all six classification code systems used in the UCO dataset. Gate 530 uses this crosswalk to handle queries where the industry context is expressed in a non-NAICS code system:

Code System	Full Name	Primary Use Context	Resolution Target
CIP	Classification of Instructional Programs	Education sector; academic program classification	→ NAICS 61xxxx
SIC	Standard Industrial Classification	Legacy system tenants; pre-1997 industry data	→ NAICS via SIC bridge table

Code System	Full Name	Primary Use Context	Resolution Target
NAICS	North American Industry Classification System	Primary routing key for all UCO node lookups	Native — primary key
SOC	Standard Occupational Classification	Labor/workforce compliance context resolution	→ NAICS via occupational-industry bridge
ISIC	International Standard Industrial Classification	International tenant contexts; multinational operators	→ NAICS via UN-US concordance table
HS / HTS	Harmonized System / Harmonized Tariff Schedule	Trade, import/export, customs compliance contexts	→ NAICS via CBP concordance table

All six code types resolve to UCO nodes via the crosswalk at query time. The resolution adds < 1ms to the L3 chain resolution step via indexed lookup on `code_crosswalk.source_code`.

6.4 NAICS-to-Agency-to-Regulation Resolution Tree

The full resolution path from an inbound tenant query to applicable UCO nodes follows this traversal sequence:

Inbound Query (with ExecutionContext.tenant_id) | ▼ [Step 1]

tenant_naics_profiles lookup | → naics_codes: ["211130", "486110"] | →

active_sectors: ["01"] // Energy | ▼ [Step 2] naics_decoder resolution | →

NAICS 211130 → "Crude Petroleum and Natural Gas Extraction" | →

primary_governing_agencies: ["BSEE/BOEM", "EPA", "FERC", "IRS"] | ▼

[Step 3] uco_nodes filter | → WHERE governing_agency IN ("BSEE/BOEM", "EPA", "FERC", "IRS") | AND (naics_code = "211130" OR is_cross_cutting = TRUE) | ▼ [Step 4] Merge with XSC layer | → UNION ALL SELECT *

FROM uco_nodes WHERE is_cross_cutting = TRUE | (19 UCO-XSC nodes always included regardless of sector filter) | ▼ [Step 5] Candidate UCO node set returned to Gate 530 | → Example: 54 ENR sector nodes + 19 XSC nodes = 73 candidates | ▼ [Gate 530] Node-by-node evaluation → aggregated gate decision

7 — Tenant NAICS Profile and Dimension Selection

7.1 Tenant Profile Structure

Each tenant's active compliance dimensions are fully determined by their row in the `tenant_naics_profiles` table (schema defined in EB Doc 3). Key fields relevant to Gate 530 dimension selection:

Field	Type	Description
<code>tenant_id</code>	UUID (PK)	Primary key — links to master tenant registry
<code>naics_codes</code>	TEXT[]	Array of registered NAICS codes for tenant's operations; supports multi-sector tenants
<code>active_sectors</code>	TEXT[]	Derived sector codes (01-19) computed from <code>naics_codes</code> at tenant onboarding
<code>include_cross_cutting</code>	BOOLEAN	Always TRUE in production; XSC layer cannot be deactivated
<code>compliance_attestations_re</code>	UUID[]	References to active

Field	Type	Description
f		compliance attestation records; used for exemption evaluation at risk weight 9+
last_naics_verified	TIMESTAMP	Date of last NAICS classification verification; alerts if stale (> 12 months)

Worked Example — Higher Education Tenant: A university with CIP code 13.xxxx (Education) maps to NAICS 611310 (Colleges, Universities, and Professional Schools) → Sector 13 (Education) → 10 UCO-EDU nodes + 19 UCO-XSC nodes = **29 total applicable compliance dimensions** evaluated at Gate 530 for every query.

7.2 Multi-Sector Tenants

Many enterprise tenants operate across multiple NAICS sectors. Gate 530 evaluates the union of all applicable nodes for every query, regardless of whether the specific query context is likely to implicate all sectors. This is a deliberate fail-safe design: the cost of evaluating additional nodes (microseconds per node) is significantly lower than the risk of missing an applicable regulation.

Worked Example — Hospital System Tenant:

- Sector 02 (Healthcare / NAICS 622110): 36 UCO-HCR nodes
- Sector 08 (Real Estate & Construction / NAICS 531210): 10 UCO-REC nodes
- Universal Cross-Cutting (UCO-XSC): 19 nodes (always included)
- **Total applicable dimensions: 65 UCO nodes evaluated per query**

Worked Example — Diversified Energy & Financial Services Conglomerate:

- Sector 01 (Energy / NAICS 211130): 54 UCO-ENR nodes
- Sector 03 (Finance & Banking / NAICS 522110): 30 UCO-FIN nodes
- Universal Cross-Cutting (UCO-XSC): 19 nodes
- **Total applicable dimensions: 103 UCO nodes evaluated per query**

7.3 Applicable Dimension Selection Algorithm

```
function resolve_applicable_nodes(tenant_id: UUID, query_context: QueryContext):
UCONode[] { // Step 1: Load tenant profile (P99
< 2ms — indexed primary key lookup) const profile = await db.query( `SELECT
* FROM tenant_naics_profiles WHERE tenant_id = $1`, [tenant_id] ); // Step 2:
Load sector-specific UCO nodes for tenant's active sectors const sector_nodes =
await db.query( `SELECT * FROM uco_nodes WHERE sector_code =
ANY($1::text[]) ORDER BY risk_weight DESC`, [profile.active_sectors] ); //
Step 3: Load cross-cutting nodes (always included; cannot be filtered) const
xsc_nodes = await db.query( `SELECT * FROM uco_nodes WHERE
is_cross_cutting = TRUE ORDER BY risk_weight DESC` ); // Step 4: Build full
candidate set const candidate_nodes = [...xsc_nodes, ...sector_nodes]; // XSC
evaluated first // Step 5: Relevance filter — include node if query context overlaps
// OR if risk_weight >= BLOCK_THRESHOLD (10) — these are always
evaluated const BLOCK_THRESHOLD = 10; const relevant_nodes =
candidate_nodes.filter(node =>
query_context.segments.includes(node.business_segment) ||
query_context.entity_types.some(et => node.specific_activity.includes(et)) ||
node.risk_weight >= BLOCK_THRESHOLD ); return relevant_nodes; } // Note:
risk_weight >= BLOCK_THRESHOLD nodes are ALWAYS included in evaluation //
even if business_segment does not overlap with query context. // This ensures
absolute-block regulations are never bypassed via context mismatch.
```

8 — Cross-Cutting Regulation Layer (UCO-XSC-5xxx)

8.1 UCO-XSC-5xxx Universal Obligations

The 19 cross-cutting nodes apply to **every tenant** in production, regardless of industry sector, NAICS classification, or business size. These represent foundational federal labor, safety, tax, and privacy obligations that attach to any organization operating within US jurisdiction. They cannot be deactivated by tenant configuration.

UCO Node ID	Regulation	Governing Body	Risk Weight	Policy Action	Enforcement Type
UCO-XSC-5001	Title VII / ADA / ADEA — Anti-Discrimination in Employment	EEOC	8	BLOCK	Injunctive
UCO-XSC-5002	FLSA — Fair Labor Standards Act (Wage & Hour Compliance)	DOL-WHD	10	BLOCK	Criminal
UCO-XSC-5003	FMLA — Family & Medical Leave Act	DOL-WHD	5	APPROVE	Administrative
UCO-XSC-5004	OSHA — General Duty Clause & Injury/Illness Recordkeeping	OSHA	9	BLOCK	Criminal

UCO Node ID	Regulation	Governing Body	Risk Weight	Policy Action	Enforcement Type
UCO-XSC-5005	OFCCP — Affirmative Action Obligations (Executive Order 11246)	OFCCP	8	BLOCK	Administrative
UCO-XSC-5006	IRS — Employment Tax Compliance & Trust Fund Recovery Penalty	IRS	10	BLOCK	Criminal
UCO-XSC-5007	State Unemployment Insurance Reporting & Contribution	State Workforce Agencies	5	APPROVE	Administrative
UCO-XSC-5008	State Data Breach Notification (All 50 States + DC)	State Attorneys General	8	BLOCK	Civil Monetary Penalty
UCO-XSC-5009	ADA Title III — Public Accommodation Accessibility	DOJ	8	BLOCK	Injunctive
UCO-XSC-5010	WARN Act — Worker Adjustment and Retraining Notification (100+ employees)	DOL	7	ESCALATE	Civil Monetary Penalty
UCO-XSC-5011	ERISA — Employee	DOL-EBSA	8	BLOCK	Civil Monetary

UCO Node ID	Regulation	Governing Body	Risk Weight	Policy Action	Enforcement Type
	Benefits Plans (Pension, Health, Welfare)				Penalty
UCO-XSC-5012	I-9 — Employment Eligibility Verification	USCIS / ICE	9	BLOCK	Civil Monetary Penalty
UCO-XSC-5013	EPA Clean Air Act — General Duty Clause (Stationary Sources)	EPA	8	BLOCK	Criminal
UCO-XSC-5014	Federal Contractor Minimum Wage (Executive Order 13658)	DOL-WHD	6	APPROVE	Administrative
UCO-XSC-5015	IRC §125 — Cafeteria Plans & Flexible Benefit Arrangements	IRS	5	APPROVE	Administrative
UCO-XSC-5016	CCPA / State Privacy Laws (California + enacted state equivalents)	California AG / State AGs	8	BLOCK	Civil Monetary Penalty
UCO-XSC-5017	SAR / BSA — Anti-Money Laundering (Financial	FinCEN	10	BLOCK	Criminal

UCO Node ID	Regulation	Governing Body	Risk Weight	Policy Action	Enforcement Type
	Nexus Triggers)				
UCO-XSC-5018	Section 508 — Accessibility Standards (Federal Contractors)	GSA / DOJ	6	ESCALATE	Administrative
UCO-XSC-5019	E-Verify — Electronic Employment Eligibility System (Federal Contractors)	USCIS / DHS	7	ESCALATE	Administrative

8.2 Cross-Cutting Evaluation Sequence in Gate 530

Cross-cutting nodes are evaluated **before** sector-specific nodes in the Gate 530 evaluation sequence. This ordering reflects two architectural principles:

23. **Universal applicability** — XSC nodes cannot be excluded or modified by NAICS profile configuration. Evaluating them first establishes a baseline gate decision state before sector-specific complexity is introduced
24. **Early termination optimization** — If the XSC layer produces a BLOCK decision (e.g., via UCO-XSC-5002 FLSA or UCO-XSC-5006 IRS), the gate decision is already determined. Sector-specific node evaluation still proceeds for audit completeness, but the gate outcome cannot change from BLOCK (fail-closed aggregation)

► Opt-Out Restriction

No tenant may opt out of cross-cutting UCO node evaluation. The `include_cross_cutting` field in `tenant_naics_profiles` is always TRUE in production. Any database record with `include_cross_cutting = FALSE` is treated as a data integrity error and triggers an alert to the security team.

9 — Gate 530 Evaluation Sequence

9.1 End-to-End Evaluation Protocol

The following ten-step protocol defines the complete Gate 530 evaluation sequence for every inbound query. Steps are executed synchronously within the L4 gate budget (< 50ms P99).

Step	Operation	Detail	Target Latency
Step 1	Receive Evaluation Request	Accept inbound IPC message from L4 Gate 530 controller via Unix domain socket (interface defined in EB Doc 1); deserialize <code>ExecutionContext</code> and <code>QueryContext</code>	< 0.5ms
Step 2	Resolve Tenant NAICS Profile	Query <code>tenant_naics_profiles</code> WHERE <code>tenant_id = ExecutionContext.tenant_id</code> ; primary key lookup on btree index; retrieve <code>naics_codes</code> , <code>active_sectors</code> , <code>compliance_attestations_ref</code>	< 2ms P99

Step	Operation	Detail	Target Latency
Step 3	Evaluate Cross-Cutting Layer	Load all 19 UCO-XSC-5xxx nodes from cache (pre-warmed on service start); evaluate each node against QueryContext; build XSC partial result set (BLOCK_LIST, ESCALATE_LIST, APPROVE_LIST)	< 5ms
Step 4	Resolve Applicable Sector Nodes	Query uco_nodes WHERE sector_code = ANY(profile.active_sectors); results pre-sorted by risk_weight DESC (highest risk first)	< 5ms
Step 5	Per-Node Evaluation	For each applicable node: compute evaluation_score (0.0-1.0) based on cosine similarity of QueryContext to node's specific_activity, business_segment, and cfr_usc_citation embeddings (vectorized via PostgreSQL pgvector or in-process)	< 20ms total
Step 6	Apply Risk Weight Threshold	For each node: if evaluation_score > 0.75 AND policy_action = BLOCK → add UCO_NODE_ID to BLOCK_LIST; if evaluation_score > 0.75 AND policy_action =	< 1ms

Step	Operation	Detail	Target Latency
		ESCALATE → add to ESCALATE_LIST; else APPROVE_LIST	
Step 7	Aggregate Gate Decision	Aggregation logic: if BLOCK_LIST non-empty → GATE_DECISION = BLOCK; else if ESCALATE_LIST non-empty → GATE_DECISION = QUARANTINE; else GATE_DECISION = PASS. Any single BLOCK triggers BLOCK for entire query (fail-closed)	< 0.5ms
Step 8	Write Audit Records	Append uco_evaluation_results rows for all evaluated nodes via append-only write using audit_writer role (no UPDATE/DELETE privileges); write is asynchronous-safe — does not block gate response	Async
Step 9	Sign Gate Decision Record	JCS/RFC 8785 canonical JSON serialization of GateDecisionRecord; Ed25519 signature using signing key identified by signer_key_id (per EB Doc 2 key lifecycle); deterministic signature — no nonce	< 1ms
Step 10	Return Signed Record	Serialize signed GateDecisionReco	< 0.5ms

Step	Operation	Detail	Target Latency
		rd and return to L4 Gate 530 controller via Unix domain socket IPC; L4 propagates decision to query execution pipeline	

9.2 Performance Characteristics

The following performance specifications are production targets for Gate 530 under nominal load conditions (single-AZ, 99th percentile measured across 30-day rolling window):

Metric	Specification	Notes
Gate evaluation latency — P50	< 8ms	Median evaluation for single-sector tenant (~29–54 nodes)
Gate evaluation latency — P99	< 50ms	Maximum budget within L4 layer; fail-closed at timeout
Maximum applicable nodes (multi-sector tenant)	~200 nodes	Hypothetical maximum: all 19 sectors + XSC
UCO node evaluation throughput	> 5,000 nodes/second	Vectorized PostgreSQL query with pgvector similarity scoring
Gate decision signing latency	< 1ms	Ed25519 deterministic signature; no network round-trip for in-process signing
Tenant NAICS profile resolution	< 2ms P99	Primary key btree index on tenant_id; shared_buffers cache hit expected
Total Gate 530 budget within L4	< 50ms P99	Hard budget enforced by L4 gate controller; timeout = BLOCK

Metric	Specification	Notes
Audit record write throughput	> 10,000 records/second	Async append-only write; does not contribute to gate latency budget

10 — Gate Decision Record Schema (UCO-Enhanced)

The Gate Decision Record (GDR) is the canonical output artifact of Gate 530. Version 2.0 of the schema adds UCO-specific fields to support attribution of gate decisions to specific UCO node IDs. All fields marked as part of the cryptographic payload are included in the JCS/RFC 8785 canonical form prior to Ed25519 signing.

```
/** * GateDecisionRecord — Version 2.0 (UCO-Native) * Schema version "2.0"
identifies UCO-native records in audit systems. * Supersedes schema version "1.0"
(abstract dimension records). * * All fields except evaluation_duration_ms are
included in JCS canonical form for signing. * Defined in: IOS+ Engineering Body
Document 4, Section 10 * Signing specification: IOS+ Engineering Body Document
2 */ interface GateDecisionRecord { // — Core Identity

———— gate_decision_id: string; // UUID v4 — unique per gate evaluation
tenant_id: string; // UUID — links to tenant_naics_profiles request_id:
string; // UUID — from ExecutionContext; correlates to query log
timestamp_iso8601: string; // UTC ISO 8601 (e.g., "2026-05-
22T13:38:00Z") // — Gate Decision

———— decision: 'PASS' | 'BLOCK' | 'QUARANTINE'; // PASS → all
```

applicable nodes evaluated; no BLOCK or ESCALATE triggers // BLOCK → one or more nodes triggered BLOCK; access denied // QUARANTINE → one or more nodes triggered ESCALATE; human review required // — UCO Evaluation Summary

uco_evaluated_node_count: number; // total UCO nodes evaluated in this request uco_block_trigger_nodes: string[]; // UCO_NODE_IDs that triggered BLOCK decision // e.g., ["UCO-XSC-5002", "UCO-ENR-1001"] uco_escalate_trigger_nodes: string[]; // UCO_NODE_IDs that triggered ESCALATE/QUARANTINE uco_max_risk_weight_seen: number; // highest risk_weight among all evaluated nodes uco_enforcement_types_triggered: string[]; // enforcement_type values of trigger nodes // e.g., ["Criminal", "Civil Monetary Penalty"] // — Compliance Chain Attribution

active_naics_codes: string[]; // tenant's registered NAICS codes at evaluation time active_sector_codes: string[]; // derived sector codes (e.g., ["01", "03", "XSC"]) compliance_chain_refs: string[]; // COMPLIANCE_CHAIN_REF strings for triggered nodes // e.g., ["CIP:15.0903 → NAICS:211130 → Federal → ..."] // — Cryptographic Integrity

record_hash: string; // SHA-256 hex of JCS-canonicalized record (all fields above) signature_ed25519: string; // Ed25519 signature (hex-encoded) per EB Doc 2 §3.2 signer_key_id: string; // Public key identifier for signature verification // Format: "smepro-gate530-{env}-{keyversion}" // — Metadata (excluded from cryptographic payload)

evaluation_duration_ms: number; // total wall-clock evaluation time in milliseconds schema_version: string; // "2.0" for UCO-native records; "1.0" = legacy (retired) }

► Schema Migration Note — v1.0 to v2.0

Version 1.0 GateDecisionRecords (abstract dimension records) are preserved in the immutable audit log for historical compliance purposes. Version 2.0 records are generated for all evaluations from the UCO-native production deployment date. Systems consuming GateDecisionRecords must handle both schema_version values. The uco_block_trigger_nodes and uco_escalate_trigger_nodes fields are empty arrays in v1.0 records.

11 — Sample UCO Nodes Per Sector

The following tables provide representative UCO node examples from each sector, drawn from the actual SMEPro Universal Compliance Decoding Matrix. These examples illustrate the regulatory density and risk profile of each sector classification.

11.1 Sector 01 — Energy (UCO-ENR-1xxx, 54 nodes)

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-ENR-1001	Offshore Drilling Plan Approval & Operations	BSEE / BOEM	30 CFR Part 250	\$40,000/day civil penalty ; operations suspension	8	BLOCK	Civil Monetary Penalty

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-ENR-1039	NRC Nuclear Facility Operating License Compliance	NRC	10 CFR Part 54	\$140,000/day per violation; license revocation; criminal	9	BLOCK	License / Certificate

11.2 Sector 02 — Healthcare (UCO-HCR-2xxx, 36 nodes)

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-HCR-1055	CMS Conditions of Participation — Acute Care Hospitals	CMS	42 CFR Part 482	Medicare/Medicaid termination; license revocation; facility closure	10	BLOCK	License / Certificate
UCO-HCR-1062	DEA Controlled Substance Registration & Dispensing Compliance	DEA	21 CFR Part 1301	\$10,000–\$25,000/violation; criminal prosecution; DEA registration revocation	10	BLOCK	Criminal

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
				on			

11.3 Sector 03 — Finance & Banking (UCO-FIN-3xxx, 30 nodes)

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-FIN-1091	Community Reinvestment Act — CRA Examination & Reporting	OCC	12 CFR Part 25	Enforcement action; merger/acquisition denial; public rating disclosure	5	APPROVE	Administrative
UCO-FIN-1098	BSA / FinCEN Anti-Money Laundering — SAR Filing & CTR Requirements	FinCEN / Treasury	31 CFR Part 1020	\$1M+/day for willful violations; criminal prosecution up to 20 years	10	BLOCK	Criminal

11.4 Sector 07 — Insurance (UCO-INS-7xxx, 35 nodes)

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-INS-7001	State Insurance License — Lines of Authority & Market Conduct	State DOI	State Insurance Codes (varies by state)	\$100K+ / violation; license suspension/revocation; cease and desist	10	BLOCK	License / Certificate
UCO-INS-7019	Risk-Based Capital (RBC) Reporting — Solvency Monitoring	NAIC / State DOI	NAIC RBC Model Act	Regulatory control; rehabilitation; liquidation proceedings	9	BLOCK	Administrative

11.5 Sector 13 — Education (UCO-EDU-13xx, 10 nodes)

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-EDU-2029	FERPA — Family Educational Rights and Privacy Act	DOE	34 CFR Part 99	Termination of federal funding ; institutional accreditation risk	5	APPROVE	Administrative
UCO-EDU-2031	Title IV Program	DOE / FSA	20 USC §1094; 34 CFR	Program participation	10	BLOCK	Criminal

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
	Participation — Student Financial Aid Compliance		Part 668	ation agreement termination; \$57,500+/violation; criminal			

11.6 Cross-Cutting Layer — Selected High-Impact Nodes (UCO-XSC-5xxx)

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
UCO-XSC-5002	FLSA — Fair Labor Standards Act (Wage & Hour, Overtime, Record keeping)	DOL-WHD	29 USC §201 et seq.	Criminal penalties + back pay + liquidated damages; personal officer liability	10	BLOCK	Criminal
UCO-XSC-5017	BSA SAR / Anti-Money Laundering — Financial Nexus Universal	FinCEN	31 USC §5318; 31 CFR Part 1010	\$1M+/day willful; criminal prosecution; asset forfeiture; 20 years impriso	10	BLOCK	Criminal

UCO Node ID	Regulation / Obligation	Governing Agency	CFR / USC Citation	Penalty Structure	Risk Weight	Policy Action	Enforcement Type
	Trigger			nment			

12 — Engineering Body Cross-References

Gate 530's UCO-native architecture has dependencies across all Engineering Body documents. The following table defines the specific dependency relationships and the exact sections or artifacts consumed from each document:

Engineering Body Document	Dependency Type	Specific Artifacts / Sections Consumed by EB Doc 4
EB Doc 1 IOS+ Core Architecture	Interface Definition	Gate 530 IPC interface specification (Unix domain socket path, message framing) ExecutionContext schema (tenant_id, request_id, session context fields) YBR Layer 4 gate budget allocation (50ms P99 total budget) L3 → L4 handoff protocol and candidate UCO node list format
EB Doc 2 Cryptographic Audit Framework	Signing Protocol	Ed25519 signing specification for Gate Decision Records JCS / RFC 8785 canonical JSON serialization procedure Key lifecycle management (key rotation, signer_key_id)

Engineering Body Document	Dependency Type	Specific Artifacts / Sections Consumed by EB Doc 4
		format) Audit writer role privilege specification (append-only, no UPDATE/DELETE)
EB Doc 3 COS+ Schema (+ Amendment v1.1)	Data Schema	uco_nodes table — full schema, indexes, UCO_NODE_ID primary key uco_evaluation_results table — audit evaluation record schema tenant_naics_profiles table — tenant dimension profile schema agency_registry table — governing agency master data code_crosswalk table (§6.5) — six-system code mapping naics_decoder table — NAICS-to-agency resolution compliance_chains table — COMPLIANCE_CHAIN_REF storage Amendment v1.1: UCO node attribute additions; is_cross_cutting flag; risk_weight floor enforcement
EB Doc 5 RAG Vault Architecture	Data Partitioning	RAG Vault sector partitions aligned with UCO sector codes (01-19 + XSC) UCO node vector embeddings — specific_activity and business_segment embedding generation Cosine similarity threshold configuration for evaluation_score computation (Step 5) Sector partition routing for query context pre-filtering
EB Doc 6 Deployment Runbook	Operational Procedure	UCO data seed scripts — initial population of all 350 nodes on fresh

Engineering Body Document	Dependency Type	Specific Artifacts / Sections Consumed by EB Doc 4
		deployment Node validation procedures — verification of 350-node count and sector distribution Cross-cutting node integrity checks — validation that 19 UCO-XSC nodes are present and active Post-deployment Gate 530 smoke tests — evaluation sequence verification UCO graph update procedures — regulatory content refresh without downtime

► Document Dependency Note

EB Doc 3 (including Amendment v1.1) is the primary data dependency for Gate 530 runtime operation. The `uco_nodes` table defined therein is the authoritative source for all compliance dimension data. Any change to UCO node content, attribute structure, or index configuration in EB Doc 3 may impact Gate 530 evaluation behavior and requires cross-document review.

12.1 Dependency Diagram — Narrative Description

The dependency flow for Gate 530's UCO-native evaluation may be described as follows: EB Doc 1 defines *how* Gate 530 receives queries (IPC interface). EB Doc 3 defines *what* Gate 530 evaluates against (UCO node schema and data). EB Doc 2 defines *how* Gate 530 signs its output (cryptographic integrity). EB Doc 5 defines *how* query context relevance is computed (vector similarity). EB Doc 6 defines *how* Gate 530 is initialized and maintained (deployment operations). All five upstream documents are required for a complete, production-grade Gate 530 implementation.

12.2 Change Control

Modifications to Gate 530's UCO evaluation logic, risk weight thresholds, policy action mappings, or node inclusion criteria constitute Architecture-Level changes under SMEPro's Engineering Body governance process. All such changes require:

- Updated entry in the affected EB Document(s) with version increment
- CCO / General Counsel sign-off (given Criminal enforcement exposure in the dataset)
- Full regression test against all 350 UCO nodes with documented evaluation results
- Updated EB Doc 6 deployment runbook entry for the change
- Re-issuance of GateDecisionRecord schema version if schema fields are modified

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Gate 530 Compliance Dimension Matrix: UCO-Native Architecture — Version 1.0 —
May 2026

Available to qualified technical evaluators under enhanced NDA only.

Unauthorized disclosure is prohibited. This document is part of the SMEPro IOS+
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